

Employee Attrition Prediction Using Machine Learning

Summary

Project Objective

The objective of this project was to identify employees who are at a higher risk of leaving the company by analyzing various factors such as job satisfaction, monthly income, work-life balance, years at the company, overtime, and performance ratings. By understanding these patterns, the HR department can take preventive actions to improve employee retention, reduce hiring costs, and maintain workforce stability.

Dataset Overview

The analysis was performed using the IBM HR Analytics Employee Attrition dataset, which contains information about **1,470 employees** and **35 employee-related attributes**. The dataset includes demographic details, job roles, salary information, work experience, satisfaction levels, and employment history. The target variable indicates whether an employee left the company or remained employed.

Key Findings

The analysis showed that employee attrition is relatively low compared to the number of employees who stay with the company, indicating an imbalanced dataset. Employees working in specific departments and job roles experienced higher attrition rates than others. Employees with lower job satisfaction, lower monthly income, frequent overtime, and fewer years at the company were more likely to leave. The first few years of employment showed the highest number of employee exits, highlighting the importance of early retention strategies.

Top Factors Influencing Employee Attrition

The most significant factors associated with employee attrition were:

- Job Satisfaction
- Overtime
- Monthly Income
- Years at the Company
- Work-Life Balance
- Total Working Years
- Job Role
- Age
- Environment Satisfaction
- Distance from Home

These factors collectively provide valuable insights into employee behavior and can help HR teams identify employees who may require additional support.

Business Recommendations

Based on the findings, the following recommendations are suggested:

1. Focus retention programs on employees during their first five years of employment through mentorship, onboarding support, and career development opportunities.
2. Improve work-life balance by reviewing overtime policies and encouraging flexible work arrangements where possible.
3. Conduct regular employee satisfaction surveys and stay interviews to identify concerns before employees decide to leave.
4. Review compensation and career growth opportunities for departments and job roles experiencing higher attrition.

5. Develop personalized career progression plans for high-performing employees to increase long-term engagement.

Conclusion

This project demonstrates how data-driven analysis can help organizations better understand employee attrition and proactively reduce employee turnover. Rather than reacting after employees resign, HR teams can use these insights to identify potential risks early and implement targeted retention strategies. Although the predictions provide valuable guidance, they should be used alongside managerial judgment and regular employee interactions to support informed decision-making.